

a lot (24-48) of ports. So it is economical to have one router and then one switch which can effectively handle 48 wired computers than have 7 routers

to do the same job. Also switches can't perform NAT (Network Address Translation) which is needed to figure out which packet is directed to which computer. There are a few more differences at the protocol level but nothing the average user needs to worry about.



Range extenders

The problem with WiFi is that it is often limited in range (30 meters) also those at the periphery of the WiFi range suffer from frequent drop in signal strength and accordingly speed. There is where range extenders or repeaters are used. These devices simply take the WiFi signal, and rebroadcast it at full strength. This dramatically increases the range and ensures less complexity of the network. Medium businesses often use Range extenders, even folks with big apartments use range extenders. Another use is to eliminate dead spots, WiFi transmitted on the 2.4 GHz band is capable of going through walls but when multiple walls or obstacles are encountered then signal strength is lost. Range extenders can boost the signal in these situations.



WiFi dongle

A WiFi dongle is an USB device which allows access to the internet via USB. This is helpful if your existing hardware is indisposed with some other function and you need to connect to the internet. Also people with old hardware tend to use them to gain the extra speed offered by newer 802.11 protocol. A GSM/CDMA dongle is similar except it uses mobile telephony to get access to the internet.

Network Cabling

The cables used to connect two or more networking devices look pretty much the same but there exists a pretty rigid standard for these. Straying from the standard doesn't mean your connection won't work but you can be sure that there will be a few kinks along the way.